ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

OPERATING SYSTEMS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 2nd year, semester 2 final examination. Section A: objective questions (10) · Section B: processes and threads, CPU scheduling, synchronisation and deadlock, memory management, file systems and disk scheduling (90).

Written by Academic Central for practice, following second-year Computer Science syllabuses at South African universities (operating systems). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Show ALL your working, including Gantt charts and tables where asked.
3. Non-programmable calculators may be used.
4. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 The kernel is …

A the core of the operating system that manages the hardware and resources.
B the graphical user interface.
C a program that compiles code.
D the BIOS chip on the motherboard.

1.2 A process that is waiting for an I/O operation to finish is in the … state.

A ready
B running
C blocked (waiting)
D terminated

1.3 Which scheduling algorithm can cause starvation of long processes?

A First-come, first-served
B Round robin
C Shortest job first
D None of these

1.4 During a context switch, the operating system saves …

A the state of the current process (registers, program counter) in its PCB.
B the whole hard disk.
C only the screen contents.
D nothing, because processes share registers.

1.5 Threads of the same process share …

A the same address space (code, data and open files).
B the same stack.
C the same program counter.
D nothing at all.

1.6 Which of the following is one of the necessary conditions for deadlock?

A Pre-emption
B Circular wait
C Paging
D Round robin

1.7 A page fault happens when a process accesses a page that is …

A not in main memory.
B in the cache.
C read-only.
D shared with another process.

1.8 Thrashing means that the system …

A spends more time swapping pages than running processes.
B has too much free memory.
C uses only one CPU core.
D formats the disk.

1.9 A semaphore is mainly used for …

A synchronisation.
B file compression.
C disk scheduling.
D encryption.

1.10 In which file allocation method does each block of a file point to the next block?

A Contiguous
B Linked
C Indexed
D Hashed

[10]

SECTION B

QUESTION 2: PROCESSES AND THREADS

2.1 Explain the difference between a program and a process. (2)

2.2 Draw the five-state process model and label every transition. (5)

2.3 Name THREE items stored in a process control block (PCB). (3)

2.4 Give TWO advantages of using several threads instead of several processes. (2)

2.5 What is a system call? Give ONE example. (3)

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QUESTION 3: CPU SCHEDULING

Four processes arrive as shown. Times are in milliseconds.

Process	Arrival time	Burst time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

3.1 Draw a Gantt chart for first-come, first-served (FCFS) scheduling and calculate the average waiting time. (5)

3.2 Draw a Gantt chart for non-pre-emptive shortest job first (SJF) scheduling and calculate the average waiting time. (5)

3.3 Draw a Gantt chart for round robin scheduling with a time quantum of 3 ms and calculate the average waiting time. (If a new process arrives at the same moment as the running process is pre-empted, the new process joins the ready queue first.) (6)

3.4 Which of the three gives the lowest average waiting time? Why is that algorithm difficult to use in a real operating system? (2)

3.5 Explain the difference between pre-emptive and non-pre-emptive scheduling. (2)

[20]

QUESTION 4: SYNCHRONISATION AND DEADLOCK

4.1 What is a race condition? Use two threads that both add 1 to a shared counter as an example. (3)

4.2 Name the THREE requirements that a solution to the critical-section problem must meet. (3)

4.3 Write pseudocode for a producer and a consumer that share a buffer of N slots. Use the semaphores empty (starts at N), full (starts at 0) and mutex (starts at 1). (6)

4.4 Name the FOUR necessary conditions for deadlock. (4)

4.5 A system has two resource types, X and Y. Available = (2, 1). Use the banker's algorithm to show whether the state is safe. If it is, give a safe sequence. (4)

Process	Allocation (X, Y)	Maximum (X, Y)
P0	(1, 0)	(3, 2)
P1	(2, 1)	(3, 2)
P2	(1, 1)	(4, 2)

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QUESTION 5: MEMORY MANAGEMENT

5.1 Explain the difference between internal and external fragmentation. (2)

5.2 A system uses 16-bit logical addresses and pages of 1 KB (1 024 bytes). How many bits are used for the page number and for the offset, and how many pages can a process have? (3)

5.3 The page table of a process maps page 0 to frame 5, page 1 to frame 2, page 2 to frame 7 and page 3 to frame 0. The page size is 1 024 bytes. Translate the logical address 2 100 into a physical address. Show your working. (3)

5.4 A process has 3 frames, which are empty at the start. It uses the pages in this order: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3 Show the frames after each reference and count the page faults for: (2 × 4) (8)

5.4.1 FIFO (first in, first out) replacement;

5.4.2 LRU (least recently used) replacement.

5.5 What is a translation lookaside buffer (TLB), and why is it used? (2)

5.6 What is Belady's anomaly? (2)

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QUESTION 6: FILE SYSTEMS AND DISK SCHEDULING

6.1 Give ONE advantage and ONE disadvantage of each file allocation method: contiguous, linked and indexed. (6)

6.2 A disk has cylinders 0 to 199. The head is at cylinder 50 and the queue of requests is: 82, 170, 43, 140, 24, 16, 190. Give the order in which the requests are served and the total head movement for: (2 × 3) (6)

6.2.1 FCFS;

6.2.2 SSTF (shortest seek time first).

6.3 Name THREE items of information that a file system keeps about each file (for example in an inode or file control block). (3)

[15]

TOTAL: 100

END OF QUESTION PAPER